Dynamical-Spacetime Discrete-Ordinates Radiation in AthenaK

Working implementation notes

April 26, 2026

1 Scope

This note documents the standalone `dyn_radiation` solver. The solver is intentionally separate from the existing `radiation` module. The legacy module assumes a stationary Cartesian Kerr-Schild (CKS) metric and stores the covariant Killing-energy form of the intensity. The new module has a CKS compatibility mode and an ADM mode intended for dynamical metric backgrounds such as the superposed boosted Kerr-Schild binary in `dynbbh.cpp`.

The first ADM implementation is deliberately conservative:

- spatial transport uses ADM Valencia coordinate speeds,
- the evolved variable is the Eulerian densitized intensity,
- an explicit ADM geometric energy source reproduces the Valencia energy source after summing over angles,
- full angular geodesic bending on a time-dependent ADM metric is not yet enabled, so ADM mode requires `angular_fluxes=false`.

2 Legacy Stationary-Metric Solver

The existing solver discretizes the grey specific intensity on angular bins n of a geodesic sphere. In a local orthonormal tetrad,

$$q_n^{(\hat{\alpha})} = (1, \ell_n, m_n, p_n), \qquad \ell_n^2 + m_n^2 + p_n^2 = 1. \tag{1}$$

The analytic CKS tetrad $e_{(\hat{\alpha})}^{\ \mu}$ gives the coordinate null direction

$$k_n^\mu = e_{(\hat{\alpha})}^{\ \mu} q_n^{(\hat{\alpha})}, \qquad k_{\mu,n} = g_{\mu\nu} k_n^\nu. \tag{2}$$

For the stationary metric, the conserved variable is

$$U_n = k_n^0 k_{0,n} I_n. \tag{3}$$

The primitive reconstructed for spatial fluxes is $k_{0,n} I_n = U_n / k_n^0$, and the face flux in direction i is

$$F_n^i = k_n^i k_{0,n} I_n. \tag{4}$$

In grey form, the finite-volume update is

$$\partial_t U_n + \partial_i F_n^i + \nabla_\Omega \cdot A_n = S_n, \tag{5}$$

where A_n is the angular flux generated by Ricci rotation coefficients. Because the CKS metric and tetrad are stationary, the tetrad, face direction maps, and angular drift coefficients are precomputed once.

3 ADM Radiation Variables

In ADM variables,

$$\mathrm{d}s^2 = -\alpha^2\mathrm{d}t^2 + \gamma_{ij}(\mathrm{d}x^i + \beta^i\mathrm{d}t)(\mathrm{d}x^j + \beta^j\mathrm{d}t), \tag{6}$$

with Eulerian normal

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta^i}{\alpha}\right). \tag{7}$$

The ADM radiation tetrad uses the Eulerian observer as its timelike leg:

$$e_{(\hat{0})}^{\ \mu} = n^\mu, \qquad e_{(\hat{a})}^{\ \mu} = (0, E_{(\hat{a})}^{\ i}), \qquad \gamma_{ij}E_{(\hat{a})}^{\ i}E_{(\hat{b})}^{\ j} = \delta_{\hat{a}\hat{b}}. \tag{8}$$

For an angular-bin direction $\ell_n^{(\hat{a})}$, define

$$s_n^i = E_{(\hat{a})}^{\ i}\ell_n^{(\hat{a})}, \qquad \gamma_{ij}s_n^i s_n^j = 1. \tag{9}$$

Photons move through the grid with coordinate velocity

$$v_n^i = \frac{\mathrm{d}x^i}{\mathrm{d}t} = \alpha s_n^i - \beta^i. \tag{10}$$

The ADM solver evolves

$$U_n = \sqrt{\gamma}\, I_n, \tag{11}$$

with spatial flux

$$F_n^i = \sqrt{\gamma}\, v_n^i I_n. \tag{12}$$

In flat space, $\sqrt{\gamma} = 1$, $\alpha = 1$, and $\beta^i = 0$, so Equations (11)–(12) reduce to ordinary transport of the tetrad-frame intensity.

4 Valencia Moment Consistency

The Eulerian radiation moments represented by the angular solution are

$$E = \sum_n w_n I_n, \tag{13}$$

$$F^i = \sum_n w_n I_n s_n^i, \tag{14}$$

$$P^{ij} = \sum_n w_n I_n s_n^i s_n^j, \tag{15}$$

where w_n is the angular quadrature weight. The Valencia energy equation in the absence of matter coupling is

$$\partial_t(\sqrt{\gamma}E) + \partial_i\left[\sqrt{\gamma}(\alpha F^i - \beta^i E)\right] = \sqrt{\gamma}\left(\alpha P^{ij}K_{ij} - F^i\partial_i\alpha\right). \tag{16}$$

The per-angle source used in `dyn_radiation` is

$$S_n^{\mathrm{geom}} = U_n\left(\alpha K_{ij}s_n^i s_n^j - s_n^i\partial_i\alpha\right). \tag{17}$$

Summing Equation (17) over angles gives exactly the right-hand side of Equation (16). This is the main reason the ADM branch does not need explicit time derivatives of gauge fields: the normal-time metric evolution is represented by K_{ij}, as in Valencia GRMHD.

The corresponding momentum equation is

$$\partial_t(\sqrt{\gamma}S_j) + \partial_i\left[\sqrt{\gamma}(\alpha P^i{}_j - \beta^i S_j)\right] = \sqrt{\gamma}\left[\frac{1}{2}\alpha P^{ik}\partial_j\gamma_{ik} + S_i\partial_j\beta^i - E\partial_j\alpha\right]. \tag{18}$$

The present ADM transport does not yet include the angular drift operator needed to reproduce this momentum source at the discrete angular level. This is why `angular_fluxes=false` is enforced for ADM mode.

5 Algorithm

At each explicit stage, ADM mode performs:

1. update or fill ADM variables;
2. construct the Eulerian tetrad by a Cholesky factorization of γ_{ij};
3. fill cell-centered $\sqrt{\gamma}$ and face-centered $\sqrt{\gamma}$, α, β^i, and face tetrad columns;
4. reconstruct $I_n = U_n/\sqrt{\gamma}$ in each spatial direction;
5. compute face fluxes with $v_n^i = \alpha s_n^i - \beta^i$;
6. update U_n with flux divergences;
7. apply Equation (17);
8. apply radiation source terms, boundary exchange, and AMR operations.

The CKS compatibility path in `dyn_radiation` keeps the legacy normalization in Equation (3); this lets the old and new solvers be compared directly on fixed Kerr-Schild backgrounds.

6 Radiation Source and Output Normalizations

The beam source uses the active radiation tetrad to build the local metric and the beam angle. It then writes the appropriate conserved variable:

$$\Delta U_n = \begin{cases} k_n^0 k_{0,n}\,\dot{I}\,\Delta t, & \text{CKS/legacy normalization,} \\ \sqrt{\gamma}\,\dot{I}\,\Delta t, & \text{ADM normalization.} \end{cases} \tag{19}$$

Radiation-moment output similarly reconstructs

$$I_n = \begin{cases} U_n/(k_n^0 k_{0,n}), & \text{CKS/legacy normalization,} \\ U_n/\sqrt{\gamma}, & \text{ADM normalization.} \end{cases} \tag{20}$$

This distinction is necessary for the flat ADM beam test to match the flat CKS beam test component by component.

7 Null Particle Beam-Edge Tracers

The particle module now has an opt-in `null_geodesic` pusher. Particle real data are interpreted as

$$(x, y, z, v_x, v_y, v_z) \quad \to \quad (x^i, k_i), \tag{21}$$

where k_i is the covariant spatial photon momentum. The coordinate-time Hamiltonian is

$$\mathcal{H}(x^i, k_i) = \alpha\sqrt{\gamma^{ij}k_ik_j} - \beta^i k_i. \tag{22}$$

The pusher advances

$$\frac{\mathrm{d}x^i}{\mathrm{d}t} = \frac{\partial \mathcal{H}}{\partial k_i} = \alpha\frac{\gamma^{ij}k_j}{K} - \beta^i, \tag{23}$$

$$\frac{\mathrm{d}k_i}{\mathrm{d}t} = -\frac{\partial \mathcal{H}}{\partial x^i} = -(\partial_i\alpha)K - \frac{\alpha}{2K}(\partial_i\gamma^{jk})k_jk_k + (\partial_i\beta^j)k_j, \tag{24}$$

with $K = \sqrt{\gamma^{ij}k_ik_j}$. The current implementation uses the nearest cell's ADM fields and centered finite differences. This is adequate for beam-edge diagnostics; higher-order interpolation can be added later without changing the data layout.

8 Comparison With Existing Radiation

Feature	`radiation`	`dyn_radiation`
Metric source	analytic stationary CKS	CKS or ADM arrays
Conserved intensity	k^0k_0I	CKS: k^0k_0I; ADM: $\sqrt{\gamma}I$
Spatial speed	k^i	CKS: k^i; ADM: $\alpha s^i - \beta^i$
Metric source terms	stationary tetrad/angular drift	ADM Valencia energy source
Angular bending	precomputed CKS coefficients	not yet enabled in ADM mode
Matter coupling	local implicit HARM-style path	copied CKS path; ADM coupling pending
BBH background	no	yes, through ADM fields

Table 1: Current solver comparison.

9 Regression Figures

Figure 1 compares the legacy CKS beam, the standalone CKS beam, and the ADM flat-space beam. Figure 2 compares radiation-hydro linear wave error metrics. Figure 3 shows the superposed-BBH beam test and null particle beam-edge tracers.

The reproducible inputs are stored in `dyngr_radiation_paper/inputs`. The beam comparison uses `beam_radiation_cks.athinput`, `beam_dynrad_cks.athinput`, and `beam_dynrad_adm_flat.athinput`. In the current regression run the maximum absolute difference in R^{00} between the legacy CKS beam and the standalone CKS beam is 0, and the maximum absolute difference between the legacy CKS beam and flat ADM beam is also 0. The radiation-hydro linear wave inputs `lwave_radiation_cks.athinput` and `lwave_dynrad_cks.athinput` give identical final error rows:

$$\text{RMS-}L_1 = 2.400907\times10^{-6}, \quad d_{L_1} = 1.685676\times10^{-6}, \quad M^1_{L_1} = 1.380892\times10^{-6}, \quad E_{L_1} = 1.007962\times10^{-6}.$$

The binary-background smoke input `dynbbh_beam_particles.athinput` exercises the ADM beam source, the superposed boosted Kerr-Schild ADM fields, and the `null_geodesic` particle pusher.

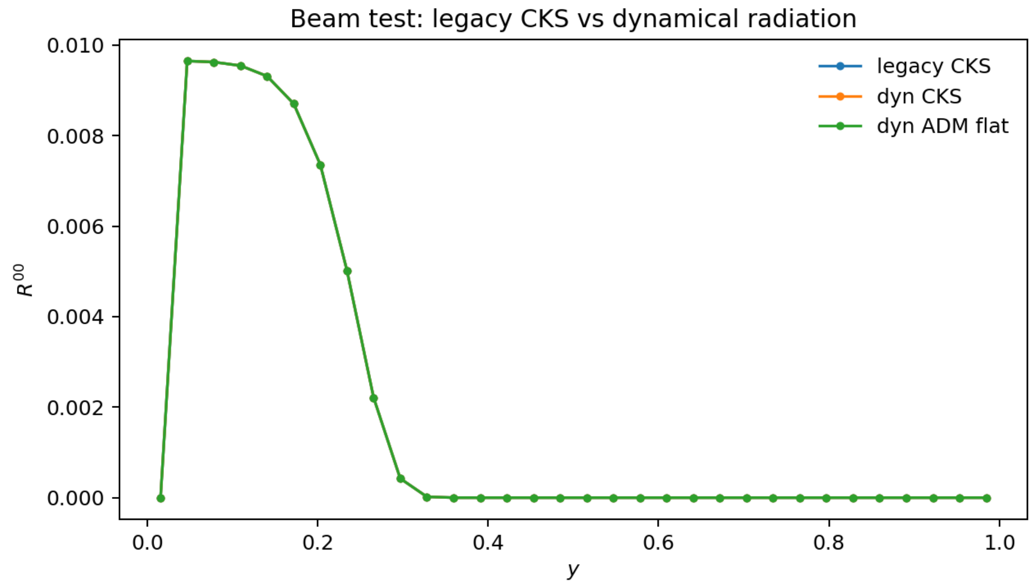

Figure 1: Beam radiation moment R^{00} along the output slice.

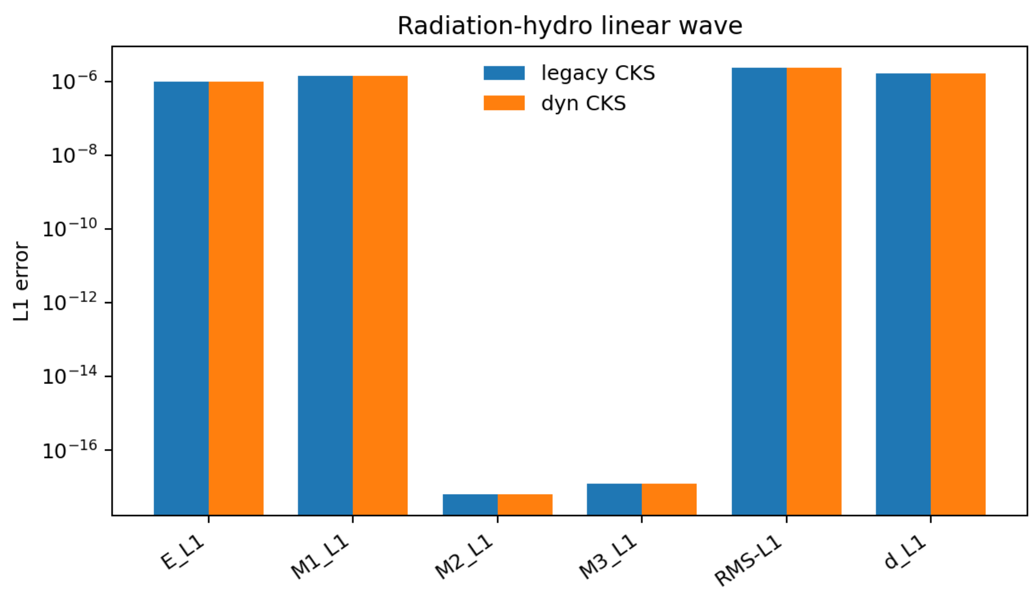

Figure 2: Linear-wave error metrics for legacy radiation and `dyn_radiation`.

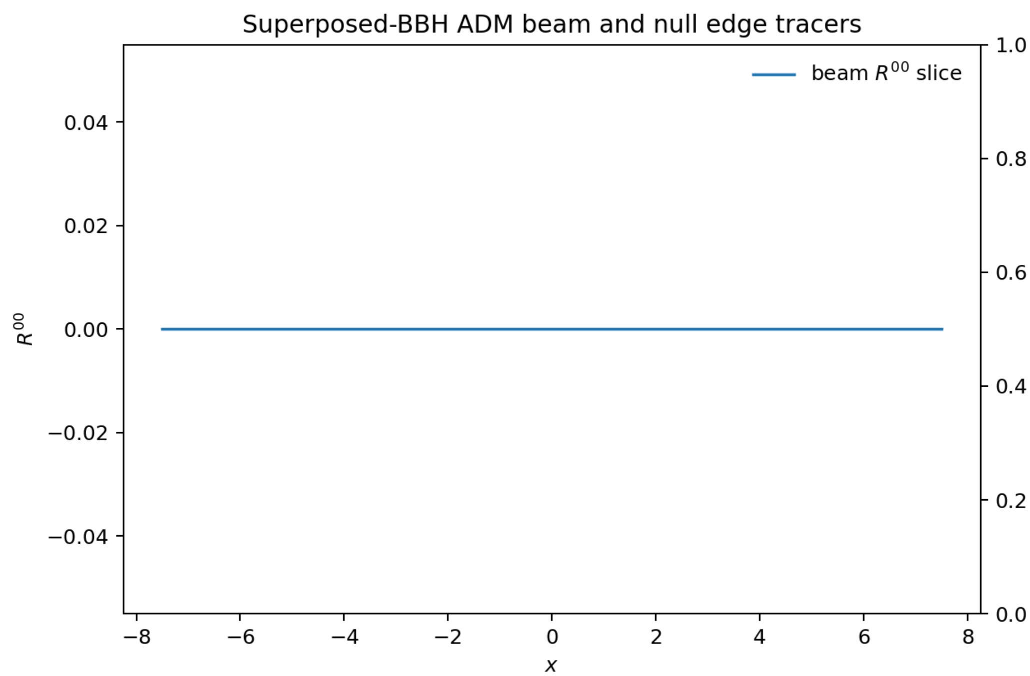

Figure 3: BBH beam slice with null particle trajectories initialized on the beam edge.